

AI-Powered Debt Relief & Financial Recovery Platform

Project Description:

The AI-Powered Debt Relief & Financial Recovery Platform is an intelligent web-based financial assistance system designed to help individuals manage their debts effectively through Artificial Intelligence and Generative AI technologies.

The platform allows users to securely register, manage multiple loan accounts, analyze their financial health, receive personalized debt settlement recommendations, and generate professional negotiation documents using Google's Gemini AI. By combining financial analytics with AI-driven insights, the system assists users in making informed financial decisions and improving their overall financial stability.

Unlike traditional budgeting or loan management applications, the proposed solution offers intelligent financial analysis, settlement probability prediction, AI-generated negotiation strategies, and personalized debt recovery recommendations based on each user's financial profile.

The platform aims to simplify debt management, reduce financial stress, and empower users with practical AI-supported recovery strategies.

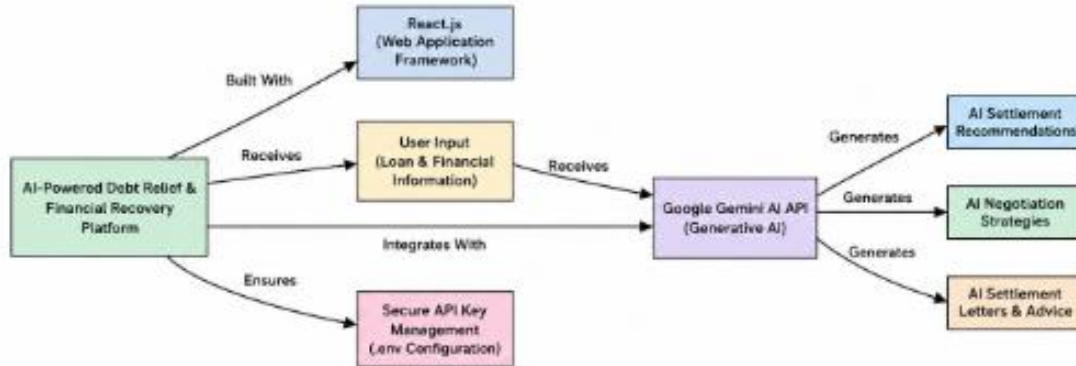
Scenarios

Scenario 1: A user has multiple loans, including personal loans, education loans, and credit card debt, making it difficult to track repayments and manage finances effectively. The platform allows the user to organize all loan information, analyze financial health, and receive personalized repayment recommendations to improve financial stability.

Scenario 2: A user experiencing financial hardship wants to negotiate with lenders but is unsure how to approach the settlement process. The platform uses Google Gemini AI to generate personalized settlement recommendations, professional settlement letters, and negotiation strategies, helping the user communicate confidently and improve the chances of successful debt resolution.

Scenario 3: A user wants to understand their financial condition and identify ways to reduce debt and improve savings. The platform analyzes income, expenses, and loan details to calculate a financial health score, identify financial risks, and provide actionable recommendations for long-term financial recovery.

Technical Architecture:



Description: The AI-Powered Debt Relief & Financial Recovery Platform follows a modular three-tier architecture to provide intelligent debt management and financial recovery services. The React.js frontend offers an interactive user interface, while the FastAPI backend processes user requests and financial data. The platform integrates with Google Gemini AI to generate settlement recommendations and negotiation strategies, with all user and loan information securely stored in an SQLite database for efficient data management.

Pre-requisites:

- Python 3.10 or above
- Node.js and npm
- React.js
- FastAPI
- SQLite Database
- Visual Studio Code
- Git and GitHub
- Google Gemini API Key
- Modern Web Browser (Google Chrome or Microsoft Edge)
- Stable Internet Connection

Project Workflow:

Milestone 1: Project Planning and Architecture

- **Activity 1.1:** Define the project objectives, problem statement, and target users for the AI-Powered Debt Relief & Financial Recovery Platform.
- **Activity 1.2:** Obtain and configure the Google Gemini API key securely using a .env file.
- **Activity 1.3:** Design the system architecture, including the React frontend, FastAPI backend, Gemini AI integration, and SQLite database.
- **Activity 1.4:** Set up the development environment by installing Python, Node.js, React, FastAPI, SQLite, and required dependencies.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the User Authentication, Loan Management, and Dashboard modules.
- **Activity 2.2:** Implement Financial Health Analysis, Debt-to-Income (DTI) calculation, and Settlement Recommendation features.
- **Activity 2.3:** Develop the backend APIs using FastAPI to process user requests and manage financial data.

Milestone 3: AI Integration

- **Activity 3.1:** Integrate Google Gemini AI to generate settlement letters, negotiation strategies, and financial advice.
- **Activity 3.2:** Store AI-generated documents and recommendations for future reference in the AI History module.
- **Activity 3.3:** Validate AI responses and ensure secure communication between the backend and Gemini AI.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop responsive user interfaces using React.js, HTML, CSS, and JavaScript.
- **Activity 4.2:** Build pages for Login, Registration, Dashboard, Loan Management, Financial Health, Settlement Recommendations, AI Negotiation, and AI History.
- **Activity 4.3:** Connect the frontend with FastAPI services to display real-time financial insights and AI-generated results.

Milestone 5: Testing and Deployment

- **Activity 5.1:** Perform functional testing, API testing, and validation of all project modules.
- **Activity 5.2:** Configure the application for deployment and verify all dependencies and environment variables.
- **Activity 5.3:** Deploy the application locally or on a cloud platform and ensure reliable access for end users.

Milestone 6: Documentation and Conclusion

- **Activity 6.1:** Prepare complete project documentation, including architecture, implementation details, testing results, and user guide.
- **Activity 6.2:** Summarize the project outcomes, challenges faced, and future enhancement opportunities.

Activity 1.1: Define the Project Objectives and Problem Statement

Before beginning development, the project's objectives, scope, and target users must be clearly identified. This helps ensure that the solution effectively addresses real-world financial challenges.

Define the Project Scope

- Identify the financial problems users face while managing multiple debts.
- Define the objectives of the AI-Powered Debt Relief & Financial Recovery Platform.
- Determine the key features such as loan management, financial health analysis, settlement recommendations, and AI-generated negotiation strategies.

Identify Target Users

- Individuals managing multiple loans.
- Users seeking debt settlement assistance.
- People requiring AI-based financial guidance.

Define Functional Requirements

- Secure user authentication.
- Loan management system.
- Financial health analysis.
- AI-powered settlement recommendations.
- AI-generated negotiation letters and financial advice.

Activity 1.2: Configure the Google Gemini API

Google Gemini AI is integrated to generate personalized settlement letters, negotiation strategies, and financial recommendations. The API key is securely configured using environment variables to ensure secure communication.

Steps

- Create a Google AI Studio account.
- Generate a Gemini API Key.
- Store the key in the .env file.
- Load the key securely using environment variables.

Activity 1.3: Design the System Architecture

A modular architecture was designed to ensure secure communication between all components of the platform. The React frontend interacts with the FastAPI backend, which processes user requests, performs financial analysis, communicates with Google Gemini AI, and stores data in the SQLite database.

Components

- React Frontend
- FastAPI Backend
- Google Gemini AI
- SQLite Database

Activity 1.4: Set Up the Development Environment

The development environment was prepared by installing the required software, configuring dependencies, and organizing the project structure.

Required Software

- Python
- Node.js

- React.js
- FastAPI
- SQLite
- Visual Studio Code
- Git

Install Dependencies

Backend: pip install -r requirements.txt

Frontend: npm install

Run the Application

Backend: uvicorn app.main:app --reload

Frontend: npm run dev

Milestone 2: Core Functionalities Development

In this milestone, the core functionalities of the **AI-Powered Debt Relief & Financial Recovery Platform** are developed. This includes implementing user authentication, loan management, financial health analysis, and debt settlement recommendations. These modules form the foundation of the platform and provide users with secure access to AI-powered financial recovery services.

Activity 2.1: Develop User Authentication and Loan Management

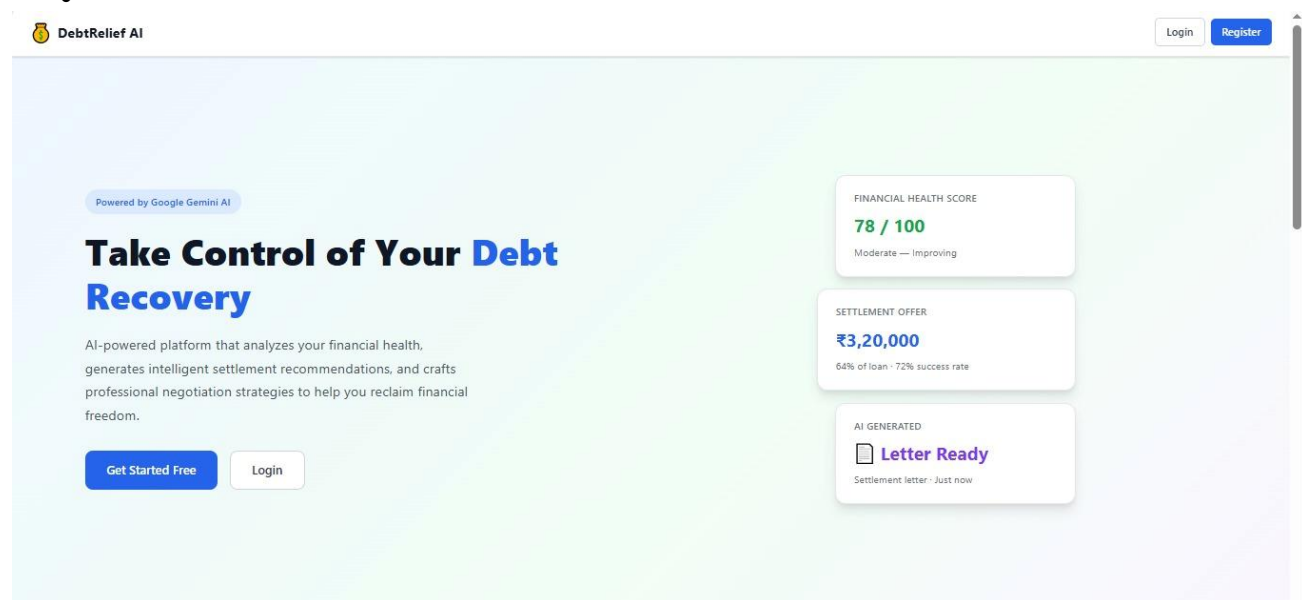
The authentication module ensures secure user registration and login, while the loan management module enables users to maintain complete records of their debts.

Implement User Authentication

- Develop secure user registration and login functionality.
- Validate user credentials before granting access.
- Maintain user sessions securely.

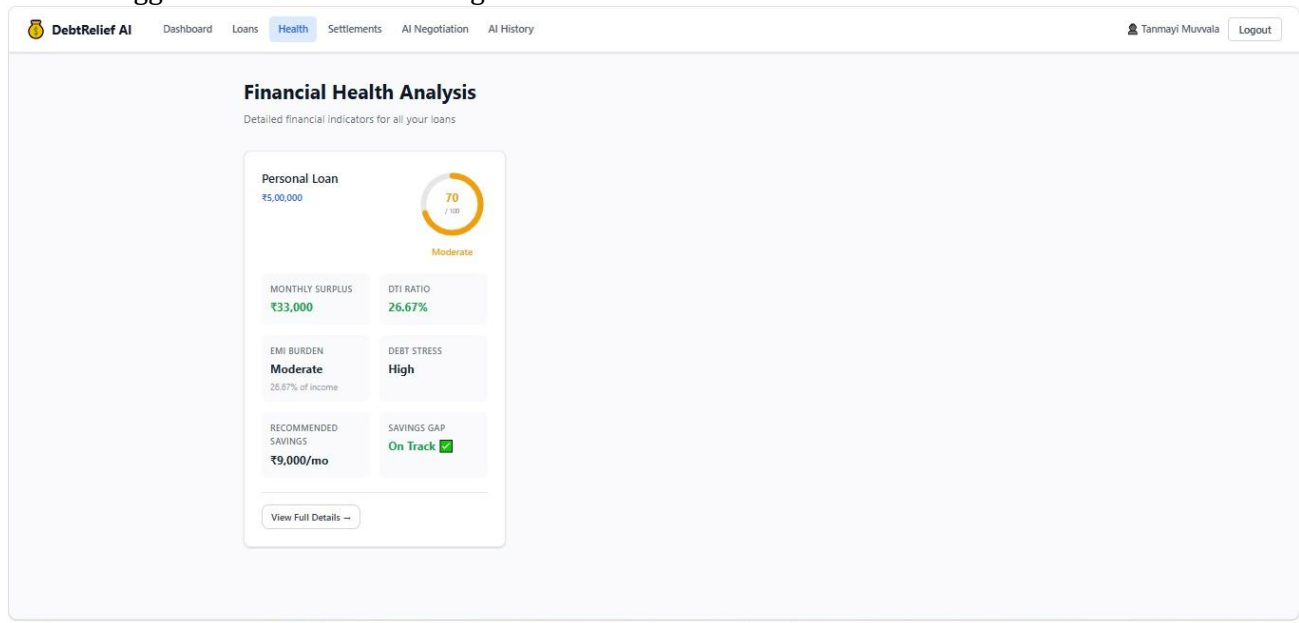
Implement Loan Management

- Add new loan records.
- View and update loan information.
- Delete loan records when required.
- Display loan summaries on the dashboard.
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Activity 2.2: Develop Financial Health Analysis

- The Financial Health Analysis module evaluates the user's financial condition by analyzing income, expenses, outstanding loans, and repayment obligations.
- **Calculate Financial Metrics**
- Calculate Debt-to-Income (DTI) Ratio.
- Calculate Financial Health Score.
- Evaluate repayment capability.
- Display financial insights graphically.
- **Generate Financial Recommendations**
- Identify financial risks.
- Recommend budget improvements.
- Suggest debt reduction strategies.



Activity 2.3: Implement Backend APIs

The backend services are developed using **FastAPI** to manage business logic and provide RESTful APIs for communication between the frontend and database.

Develop REST APIs

- User Authentication APIs
- Loan Management APIs
- Financial Analysis APIs
- Settlement Recommendation APIs
- AI History APIs

API Responsibilities

- Process user requests.
- Validate financial information.
- Retrieve and store loan records.
- Return AI-generated responses.

Milestone 3: AI Integration

This milestone focuses on integrating **Google Gemini AI** to generate intelligent financial recommendations, settlement letters, negotiation strategies, and personalized financial guidance based on the user's financial profile.

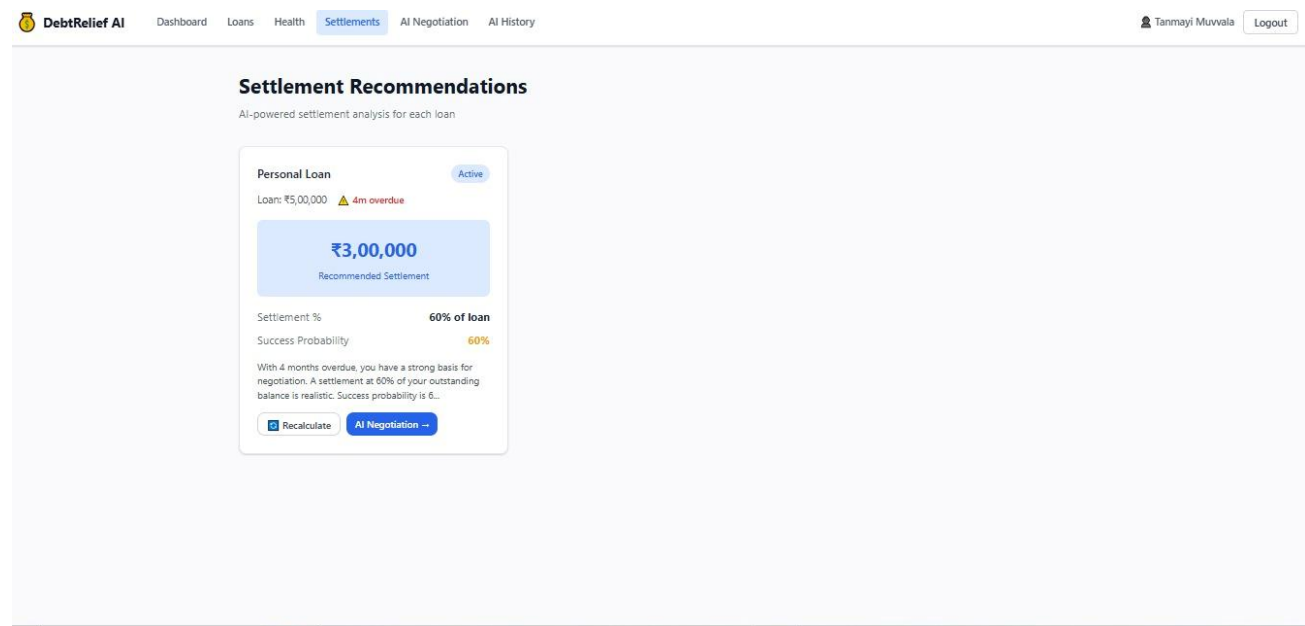
Activity 3.1: Integrate Google Gemini AI

Google Gemini AI is connected with the backend to generate intelligent financial content dynamically.

AI Features

- Settlement Recommendation
- Settlement Letter Generation
- Negotiation Strategy
- Financial Advice

The backend securely sends user financial data to Gemini AI and displays the generated response to the user.



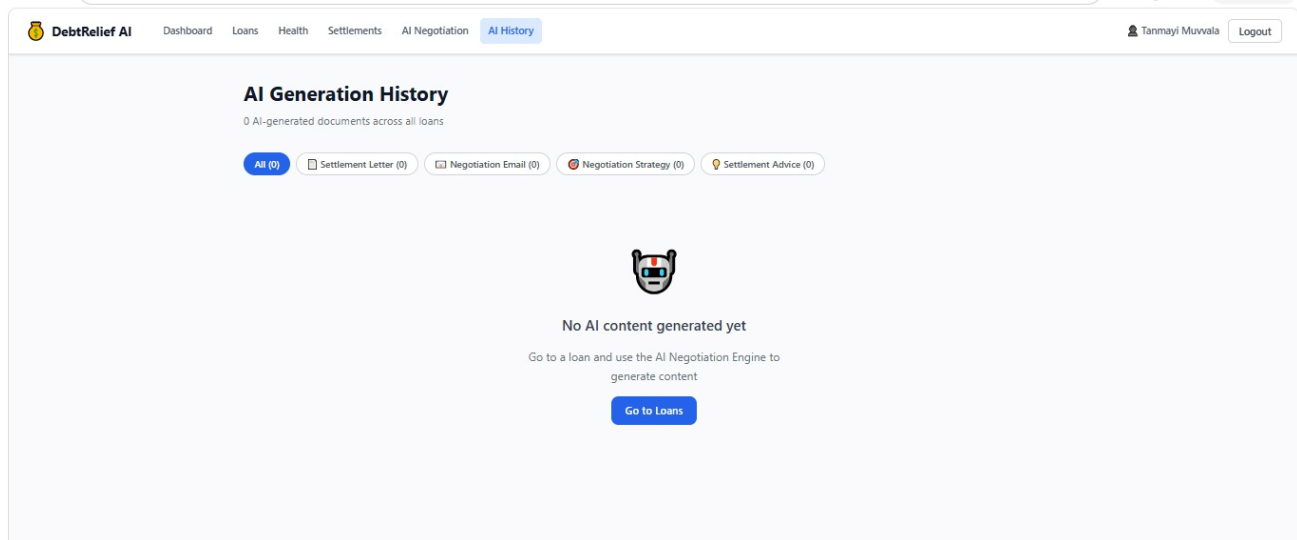
The screenshot shows the 'DebtRelief AI' dashboard with a navigation bar including 'Dashboard', 'Loans', 'Health', 'Settlements', 'AI Negotiation', and 'AI History'. The 'Settlements' tab is active. The main section is titled 'Settlement Recommendations' with the subtitle 'AI-powered settlement analysis for each loan'. A card for a 'Personal Loan' (Active) shows a loan amount of ₹5,00,000, which is 4 months overdue. The recommended settlement is ₹3,00,000, which is 60% of the loan. The success probability is 60%. A message states: 'With 4 months overdue, you have a strong basis for negotiation. A settlement at 60% of your outstanding balance is realistic. Success probability is 60%.' At the bottom of the card are buttons for 'Recalculate' and 'AI Negotiation -->'.

Activity 3.2: Store AI Generated Results

All AI-generated recommendations and documents are stored to allow users to review previous analyses.

AI History Module

- Save generated settlement recommendations.
- Save negotiation strategies.
- Save financial advice.
- Retrieve previous AI responses.



Activity 3.3: Validate AI Responses

The backend validates all responses generated by Google Gemini AI before presenting them to users.

Validation Process

- Verify response completeness.
- Handle invalid AI outputs.
- Display meaningful error messages.
- Ensure reliable AI-generated recommendations.

Milestone 4: Frontend Development

Milestone 4 focuses on developing a responsive and user-friendly interface using **React.js**. The frontend enables users to interact seamlessly with the platform while displaying AI-generated insights and financial analytics.

Activity 4.1: Design the User Interface

A modern and responsive interface is developed to provide an intuitive user experience.

UI Components

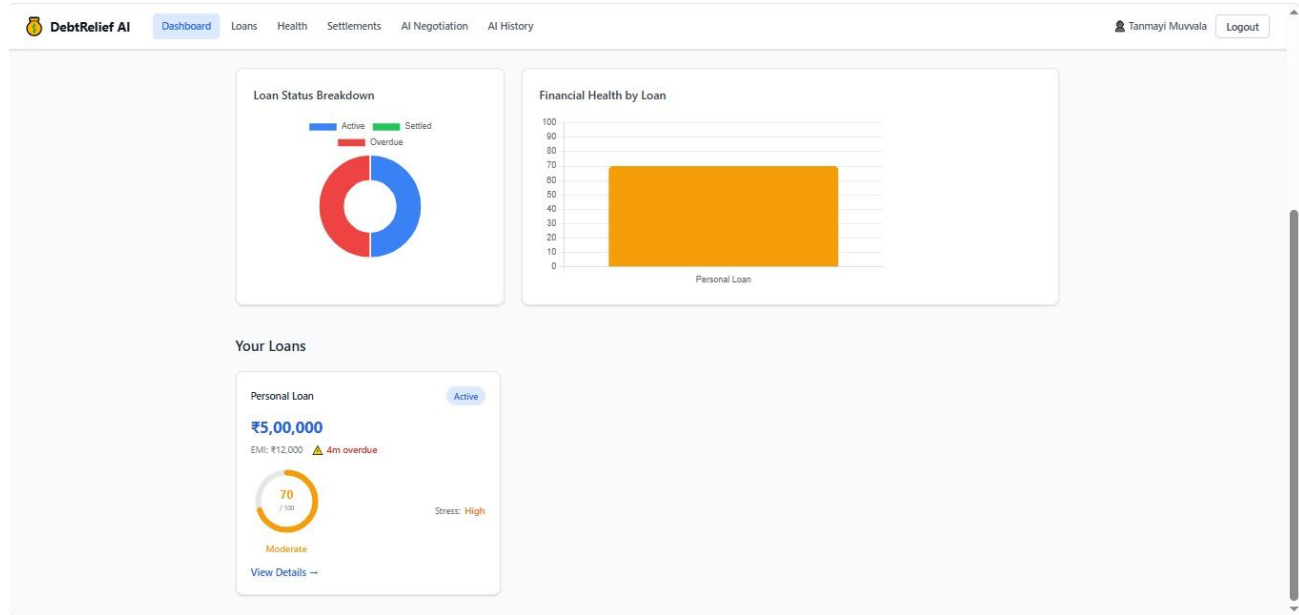
- Landing Page
- Login
- Registration
- Dashboard
- Loan Management
- Financial Health
- Settlement Recommendations
- AI History

Activity 4.2: Connect Frontend with Backend

The React frontend communicates with FastAPI through REST APIs.

Features

- Display loan information.
- Fetch financial analysis.
- Retrieve AI-generated recommendations.
- Update dashboard dynamically.



Activity 4.3: Responsive Design

The application is optimized for different screen sizes to provide a consistent user experience.

Improvements

- Responsive layouts.
- Improved navigation.
- Better accessibility.
- Faster page loading.

Milestone 5: Testing and Deployment

This milestone ensures that all modules function correctly through comprehensive testing before deployment.

Activity 5.1: Functional Testing

Test all application modules.

Tested Modules

- User Registration
- Login
- Loan Management
- Financial Health Analysis
- Settlement Recommendation
- AI History

Activity 5.2: Local Deployment

Configure the application for local execution.

Deployment Steps

- Install dependencies.
- Configure .env.
- Run FastAPI backend.
- Run React frontend.
- Verify application functionality.

Activity 5.3: Application Verification

Verify all project modules before deployment.

Verification

- User Authentication
- Loan CRUD Operations
- Financial Analysis
- AI Recommendations
- Dashboard
- AI History

 **DebtRelief AI** | [Dashboard](#) | [Loans](#) | [Health](#) | [Settlements](#) | [AI Negotiation](#) | [AI History](#)

Tanmayi Muvvala | [Logout](#)

Settlement Recommendations

AI-powered settlement analysis for each loan

Personal Loan Active

Loan: ₹5,00,000 4m overdue

₹3,00,000
Recommended Settlement

Settlement % **60% of loan**

Success Probability **60%**

With 4 months overdue, you have a strong basis for negotiation. A settlement at 60% of your outstanding balance is realistic. Success probability is 6...

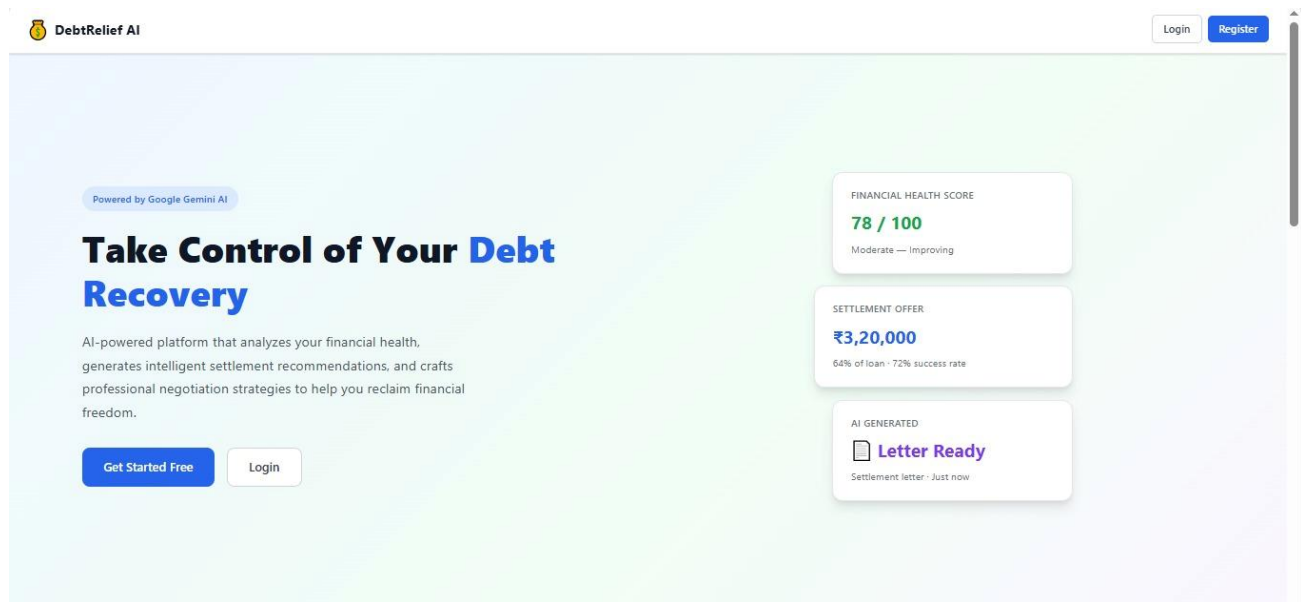
[Recalculate](#) [AI Negotiation -->](#)

Milestone 6: Conclusion

The **AI-Powered Debt Relief & Financial Recovery Platform** successfully integrates Artificial Intelligence with financial management to provide users with intelligent debt recovery solutions. The platform enables secure loan management, financial health analysis, AI-powered settlement recommendations, and negotiation assistance through Google Gemini AI. By combining modern web technologies with Generative AI, the system offers a user-friendly, scalable, and efficient solution that helps individuals improve their financial well-being.

Exploring the Website's Web Pages

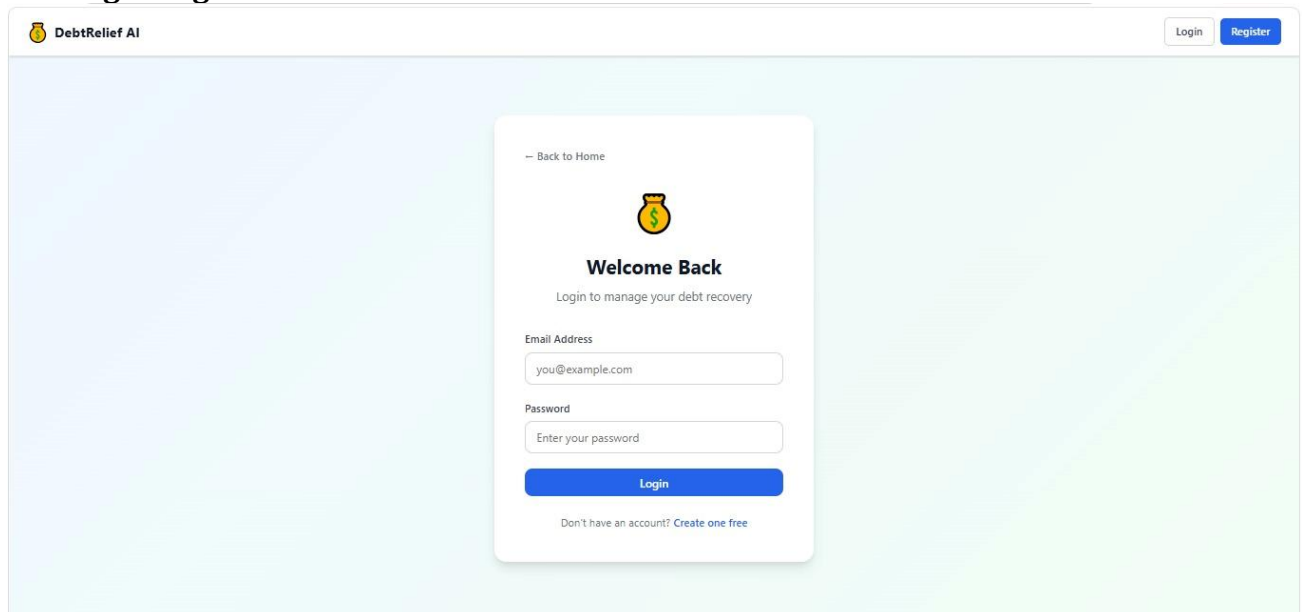
1. Landing Page



Description:

The Landing Page serves as the entry point of the AI-Powered Debt Relief & Financial Recovery Platform. It introduces the platform's purpose, highlights its key features, and provides users with options to register or log in. The clean and responsive interface offers an intuitive experience while encouraging users to begin their financial recovery journey.

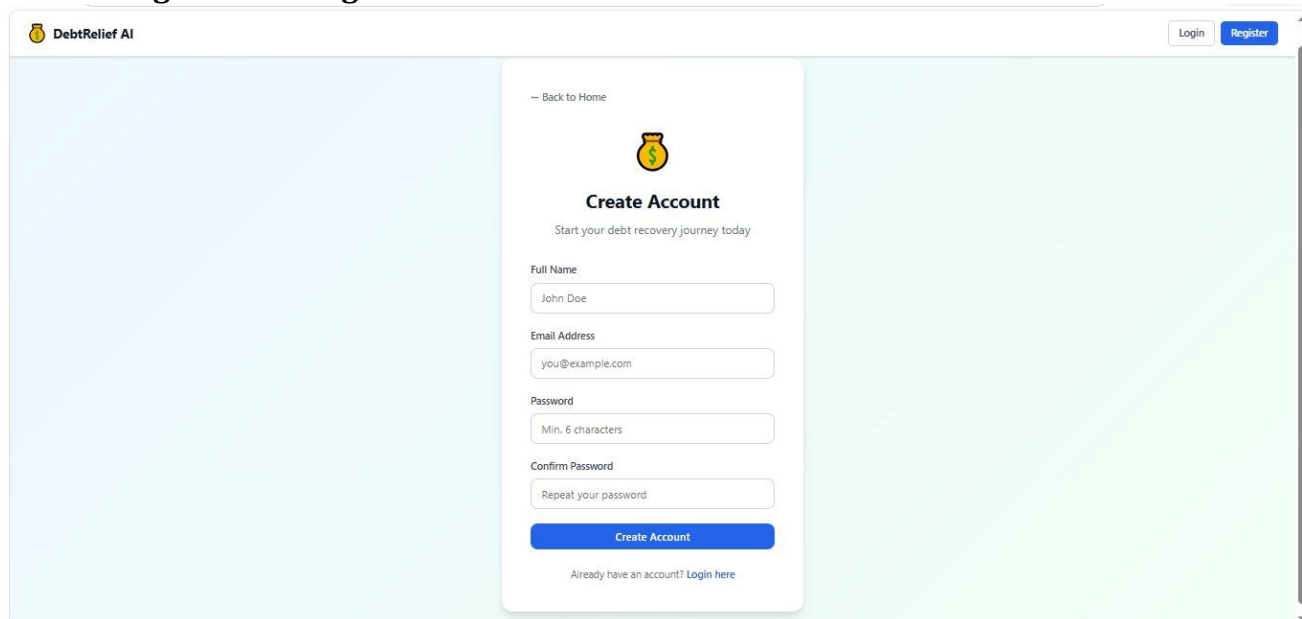
2. User Login Page



Description:

The Login Page allows registered users to securely access the platform using their email and password. User credentials are validated through the FastAPI backend, ensuring secure authentication before granting access to personalized financial information.

3. User Registration Page



DebtRelief AI

Login Register

— Back to Home



Create Account

Start your debt recovery journey today

Full Name

John Doe

Email Address

you@example.com

Password

Min. 6 characters

Confirm Password

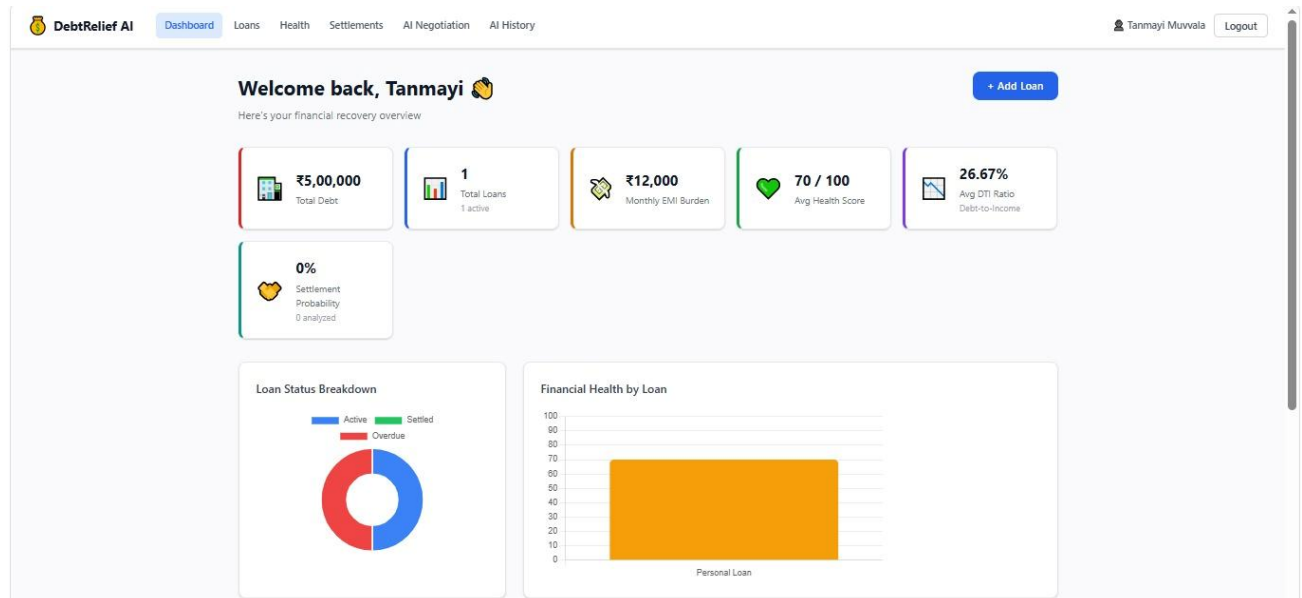
Repeat your password

Create Account

Already have an account? Login here

DESCRIPTION:The Registration Page enables new users to create an account by providing their personal details and login credentials. After successful registration, users can securely access all financial management and AI-powered services offered by the platform.

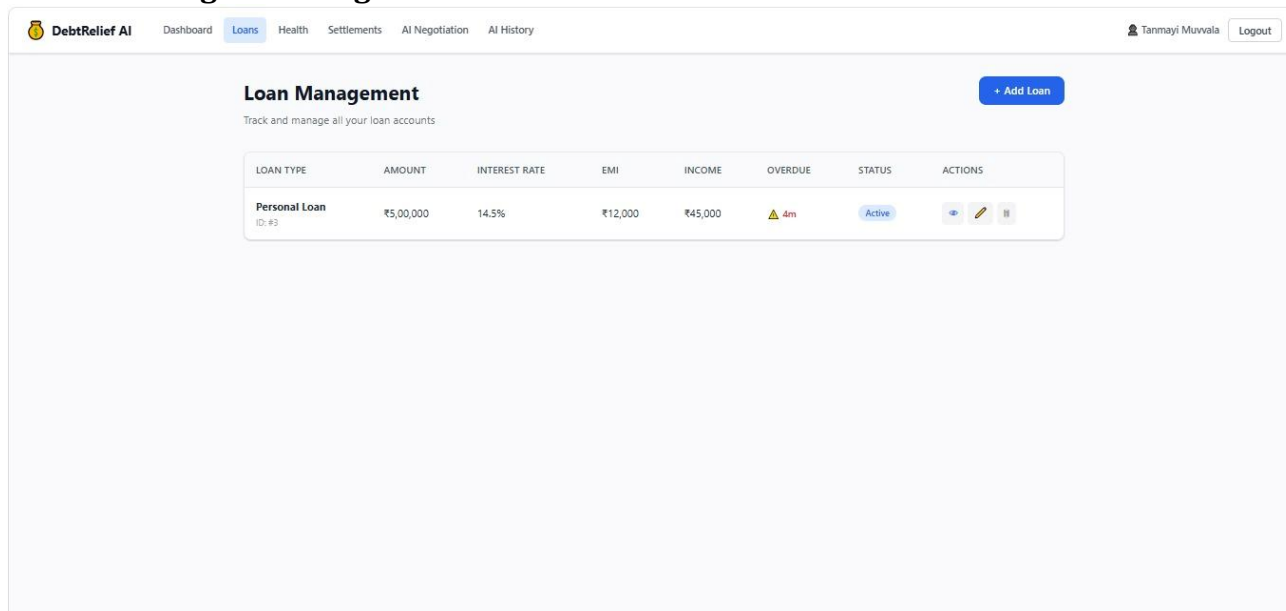
4. Dashboard



Description:

The Dashboard provides users with a comprehensive overview of their financial information. It displays loan summaries, financial statistics, repayment details, and quick access to various modules, allowing users to monitor their financial health efficiently.

5. Loan Management Page



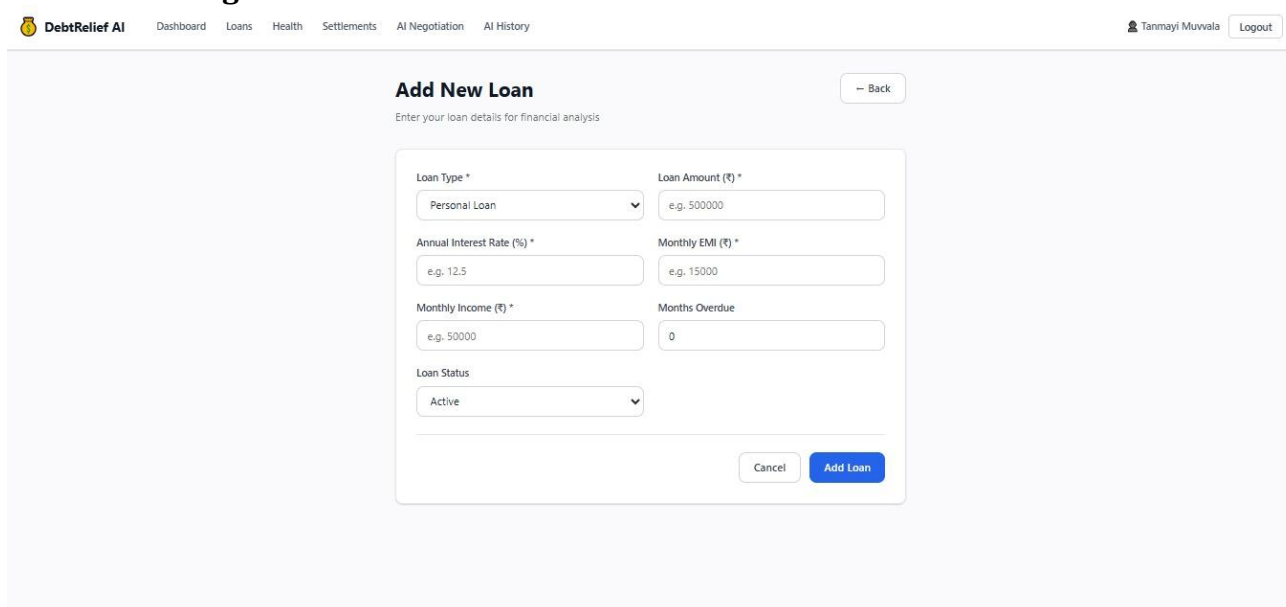
The screenshot shows the 'Loan Management' page in the DebtRelief AI dashboard. The page has a navigation bar with 'DebtRelief AI', 'Dashboard', 'Loans' (active), 'Health', 'Settlements', 'AI Negotiation', and 'AI History'. A user profile 'Tanmayi Muvvala' and a 'Logout' button are in the top right. The main content area is titled 'Loan Management' with a subtitle 'Track and manage all your loan accounts' and an '+ Add Loan' button. Below this is a table with columns: LOAN TYPE, AMOUNT, INTEREST RATE, EMI, INCOME, OVERDUE, STATUS, and ACTIONS. The table contains one row for a 'Personal Loan' with ID #3, amount ₹5,00,000, interest rate 14.5%, EMI ₹12,000, income ₹45,000, overdue 4m, status 'Active', and action icons for edit, delete, and details.

LOAN TYPE	AMOUNT	INTEREST RATE	EMI	INCOME	OVERDUE	STATUS	ACTIONS
Personal Loan ID: #3	₹5,00,000	14.5%	₹12,000	₹45,000	⚠️ 4m	Active	✎️ ✖️ ⓘ

Description:

The Loan Management Page allows users to view, add, update, and delete loan records. It maintains complete information about outstanding loans, helping users organize their financial obligations and manage repayments effectively.

6. Add Loan Page

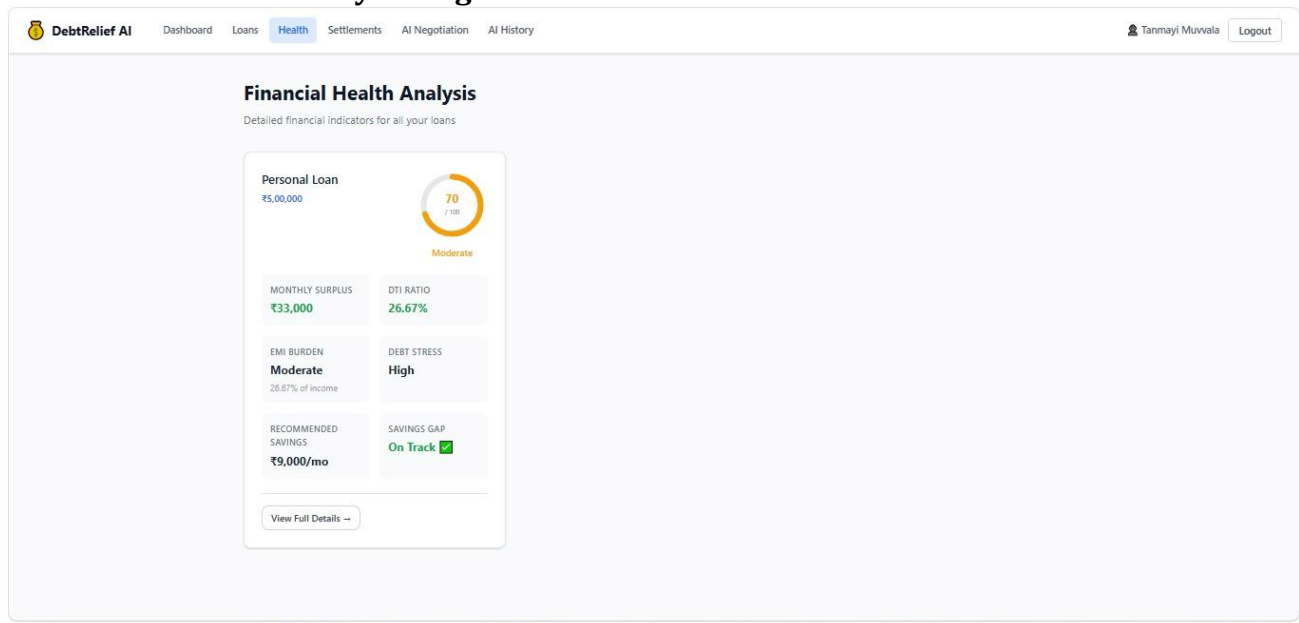


The screenshot shows the 'Add New Loan' page in the DebtRelief AI dashboard. The page has a navigation bar with 'DebtRelief AI', 'Dashboard', 'Loans' (active), 'Health', 'Settlements', 'AI Negotiation', and 'AI History'. A user profile 'Tanmayi Muvvala' and a 'Logout' button are in the top right. The main content area is titled 'Add New Loan' with a subtitle 'Enter your loan details for financial analysis' and a 'Back' button. Below this is a form with fields for: Loan Type (dropdown, 'Personal Loan'), Loan Amount (₹) (text input, 'e.g. 500000'), Annual Interest Rate (%) (text input, 'e.g. 12.5'), Monthly EMI (₹) (text input, 'e.g. 15000'), Monthly Income (₹) (text input, 'e.g. 50000'), Months Overdue (text input, '0'), and Loan Status (dropdown, 'Active'). At the bottom are 'Cancel' and 'Add Loan' buttons.

Description:

The Add Loan Page enables users to enter new loan details, including loan type, principal amount, interest rate, EMI, and repayment duration. These details are securely stored and used for financial analysis and AI-powered recommendations.

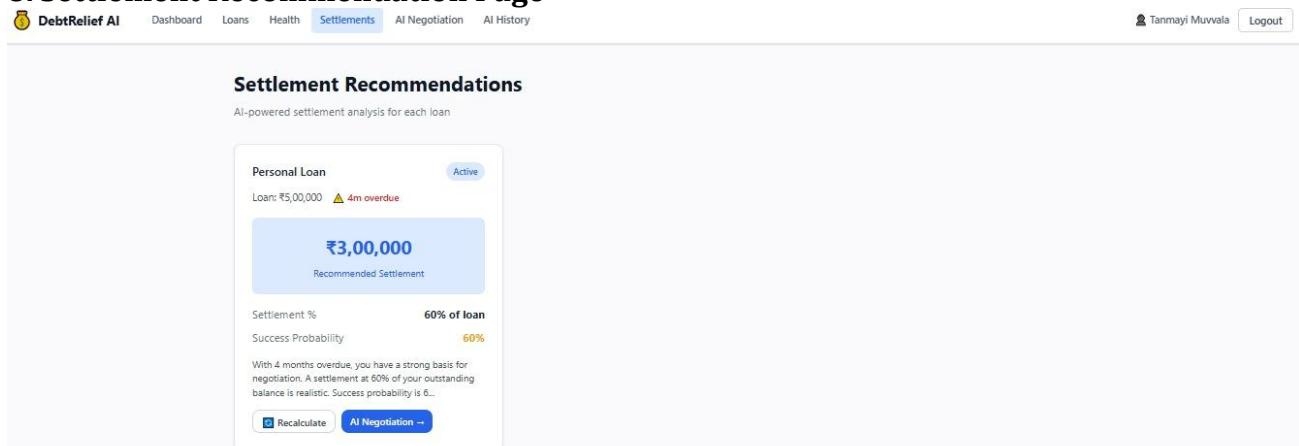
7. Financial Health Analysis Page



Description:

The Financial Health Analysis Page evaluates the user's financial condition by analyzing income, expenses, outstanding loans, and debt obligations. It calculates key financial metrics and presents insights that help users understand their financial stability and identify areas for improvement.

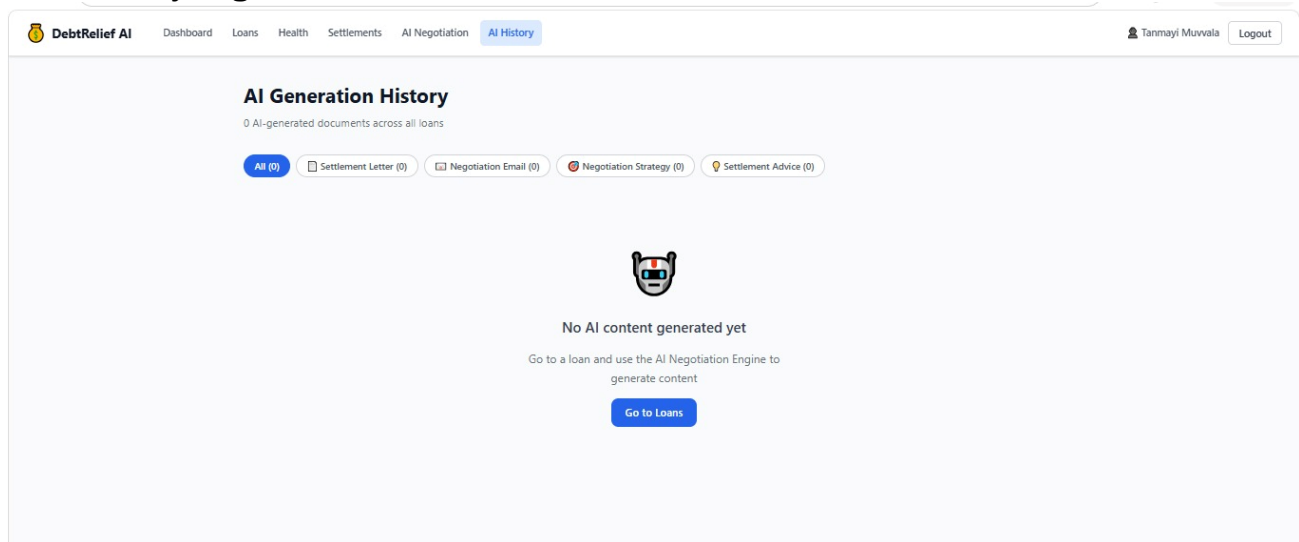
8. Settlement Recommendation Page



Description:

The Settlement Recommendation Page uses Google Gemini AI to generate personalized debt settlement suggestions based on the user's financial profile. It recommends suitable settlement amounts and provides intelligent financial guidance to assist users in reducing their debt burden.

9. AI History Page



Description:

The AI History Page stores all previously generated AI recommendations, settlement letters, and negotiation strategies. Users can revisit earlier analyses and track their financial recovery journey without generating new responses each time.

Conclusion:

The **AI-Powered Debt Relief & Financial Recovery Platform** successfully integrates Artificial Intelligence with modern web technologies to provide an intelligent solution for debt management and financial recovery. By leveraging **Google Gemini AI**, the platform delivers personalized financial health analysis, debt settlement recommendations, professional negotiation strategies, and AI-generated financial guidance based on each user's financial profile.

Developed using **React.js**, **FastAPI**, and **SQLite**, the platform offers a secure, responsive, and user-friendly environment for managing loans and monitoring financial progress. Throughout the project, key phases including system design, implementation, AI integration, testing, and deployment were successfully completed, resulting in a reliable and scalable application.

In the future, the platform can be enhanced by integrating real-time banking APIs, credit score monitoring, predictive financial analytics, multilingual support, and mobile application support. These enhancements will further strengthen the platform's ability to assist users in achieving long-term financial stability and smarter financial decision-making.